

Clarification on the adopted methodology

for RWA projections & Treasury-RWA Schemas

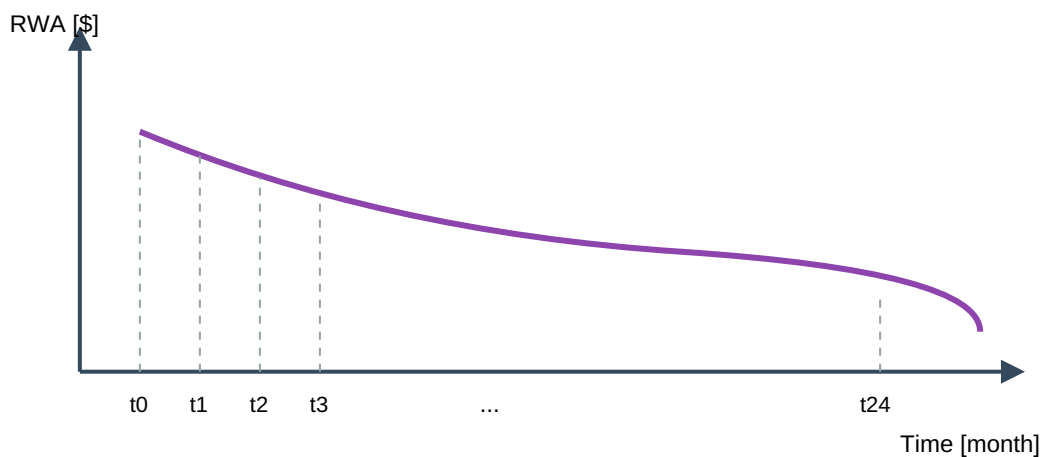
1. Execution Configuration

The module's configuration relies on a robust settings schema, leveraging modern validation to ensure data integrity during runs.

```
from datetime import date
from pydantic import Field
from pydantic_settings import BaseSettings

class RunInstructions(BaseSettings):
    run_date: date = Field(default_factory=date.today)
    projected_months: int = Field(default=24, gt=0, le=24)
```

2. How dates are calculated



Projection is calculated as follow:

- **t_0** – run date
- **t_1** – last day of the current month
- **t_2** – last day of the next month

- etc.

Only **t0** changes depending on the run execution, and the rest of the dates are always calculated on the last day of the month.

Examples of calculation

For t0 = 16/12/2025:

time	date
t0	16/12/2025
t1	31/12/2025
t2	31/01/2026
t3	28/02/2026

For t0 = 31/12/2025 (t0 and t1 would be the same):

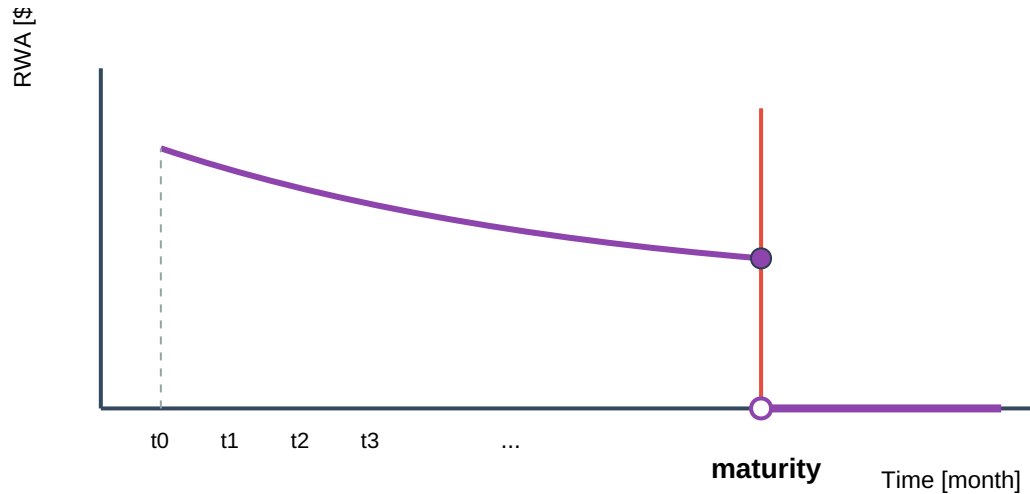
time	date
t0	31/12/2025
t1	31/12/2025
t2	31/01/2026
t3	28/02/2026

For t0 = 01/01/2026:

time	date
t0	01/01/2026
t1	31/01/2026
t2	28/02/2026
t3	31/03/2026

3. How Projection is calculated in context of maturity

We have adopted an approach that for maturity eq 0 we calculate values. For negative maturity we project 0. For none values (missing) we do not project (values also are none).



In special case when the asset realizes on the last day of the month (we have maturity zero on this day), we should see calculated value.

time	date	maturity	RWA
t0	01/01/2026	0.3	1.0
t1	31/01/2026	0.2	1.0
t2	28/02/2026	0.1	1.0
t3	31/03/2026	0.0	1.0
t4	30/04/2026	-0.1	0.0

4. Schemas produced by Treasury-RWA

For t0 it is produced **OutputSuccess**

```
import pandera as pa
from pandera.typing import Series

class OutputSuccess(pa.DataFrameModel):
    id: Series[str] = pa.Field(unique=True)
    counterparty_gid: Series[str]
    basel_3_0_pd: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_pd: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_0_dlgd: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_dlgd: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rw_final: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rwa_final: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_ro_rw_final: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_0_rw_final: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_0_rwa: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_0_ro_rw: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rw_foundation: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rwa_foundation: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_ro_rw_foundation: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rw_standardised: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_rwa_standardised: Series[float] = pa.Field(ge=0, nullable=True)
    basel_3_1_ro_rw_standardised: Series[float] = pa.Field(ge=0, nullable=True)
```

OutputProjection

For projection there is produced t0 for run_date (today) and 24 months. Where each point is calculated on the last day of the month.

```
from datetime import date
```

```
class OutputProjection(pa.DataFrameModel):  
    id: Series[str] = pa.Field(unique=False)  
    projection_date: Series[date] = pa.Field(nullable=False)  
    basel_3_0_rw_final: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_0_rwa: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_0_ro_rw: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_rw_foundation: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_rwa_foundation: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_ro_rw_foundation: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_rw_standardised: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_rwa_standardised: Series[float] = pa.Field(ge=0, nullable=True)  
    basel_3_1_ro_rw_standardised: Series[float] = pa.Field(ge=0, nullable=True)
```